

Introduction to Python

1. Which of the following is a programming language used to give instructions to a computer?

- A. Operating system
- B. Programming language
- C. Compiler
- D. Algorithm

2. A program written in a high-level language is called

- A. Object code
- B. Machine code
- C. Source code
- D. Bytecode

3. Python uses which of the following to execute programs?

- A. Compiler
- B. Interpreter
- C. Assembler
- D. Linker

4. Which of the following statements is TRUE about Python interpreter?

- A. It translates the whole program at once and then executes
- B. It processes program statements one by one
- C. It does not stop even if there is an error
- D. It converts machine language to high-level code

5. What is the difference between a compiler and an interpreter?

- A. Compiler translates program line by line; interpreter translates all at once
- B. Compiler translates all at once; interpreter translates line by line
- C. Both are the same
- D. Compiler executes the program, interpreter does not

Features of Python

6. Which of the following is NOT a feature of Python?

- A. Case-sensitive
- B. Portable
- C. Interpreted language
- D. Can run without an interpreter

7. Python uses **indentation** to

- A. Highlight keywords
- B. Separate variables
- C. Define blocks and nested blocks
- D. Increase speed of execution

8. Which of the following is TRUE about Python libraries?

- A. Python has no built-in functions
- B. Python has a rich library of predefined functions

- C. Python can only use third-party libraries
- D. Python libraries are only for web development

9. Python is considered **platform independent because**

- A. It runs only on Windows
 - B. It can run on multiple operating systems without modification
 - C. It requires no interpreter
 - D. It is compiled into machine language
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Python Execution Modes

10. Which Python mode allows execution of one statement at a time?

- A. Script mode
- B. Interactive mode
- C. Batch mode
- D. Compiler mode

11. Which Python mode allows writing multiple instructions in a file and executing it?

- A. Interactive mode
- B. Script mode
- C. Debug mode
- D. REPL mode

12. Python source code files are saved with which extension?

- A. .java
- B. .cpp
- C. .py
- D. .exe

13. Which of the following is a limitation of interactive mode?

- A. Cannot execute a single line
 - B. Cannot save statements for future use
 - C. Cannot run the Python interpreter
 - D. Cannot perform calculations
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Keywords and Identifiers

14. Python keywords are

- A. Names chosen by the programmer
- B. Reserved words with special meaning
- C. Variables
- D. Comments

15. Which of the following can be a valid identifier in Python?

- A. 1variable
- B. _studentName
- C. @marks
- D. #total

- 16.** Python identifiers cannot
- A. Start with a letter or underscore
 - B. Contain alphabets
 - C. Start with a digit
 - D. Contain numbers anywhere except first character
- 17.** Which of the following is a valid use of identifiers?
- A. Assigning values to variables
 - B. Naming functions
 - C. Naming classes
 - D. All of the above
-

Variables

- 18.** A variable in Python is
- A. A fixed memory location
 - B. An object that stores data in memory
 - C. Always numeric
 - D. Only a string
- 19.** Which of the following is TRUE about variable declaration in Python?
- A. Variables must be explicitly declared before use
 - B. Variables are automatically declared when assigned a value
 - C. Variables can be used before assignment
 - D. Python does not allow variables
- 20.** In Python, which of the following is a numeric variable?
- A. `message = "Hello"`
 - B. `userNo = 101`
 - C. `name = 'Alice'`
 - D. `greeting = "Welcome"`
-

Comments

- 21.** In Python, comments are used to
- A. Make the program execute faster
 - B. Add remarks for understanding the code
 - C. Store data
 - D. Create variables
- 22.** Python comment starts with which symbol?
- A. `//`
 - B. `/*`
 - C. `#`
 - D. `--`
- 23.** Which of the following lines is a comment in Python?
- A. `#This is a comment`
 - B. `print("Hello")`

- C. userNo = 101
- D. area = length * breadth

- 24.** Comments in Python are
- A. Executed by the interpreter
 - B. Ignored by the interpreter
 - C. Stored as variables
 - D. Converted to machine code
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Arithmetic Operators

- 25.** Which operator is used in Python for exponentiation?

- A. ^
- B. **
- C. %
- D. //

- 26.** What is the output of the following code?

```
num1 = 5  
num2 = 6  
print(num1 + num2)
```

- A. 56
- B. 11
- C. 1
- D. 30

- 27.** Which operator in Python returns the remainder after division?

- A. //
- B. %
- C. /
- D. *

- 28.** Which of the following statements is TRUE about string multiplication?

- A. Strings cannot be multiplied
- B. 'India' * 2 outputs 'IndiaIndia'
- C. 'India' * 2 outputs 'India2'
- D. Strings can only be multiplied by another string

- 29.** What does the // operator do in Python?

- A. Returns the remainder
 - B. Returns floating-point division
 - C. Returns quotient after removing the decimal part
 - D. Raises a number to power
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Relational Operators

30. What will be the output of `num1 == num2` if `num1 = 10` and `num2 = 0`?

- A. True
- B. False
- C. Error
- D. 0

31. Which relational operator checks if two values are not equal?

- A. `!=`
- B. `==`
- C. `>`
- D. `<`

32. Which of the following evaluates True if `str1 >= str2`?

- A. `str1` comes alphabetically after `str2` or is equal
- B. `str1` comes alphabetically before `str2`
- C. `str1` is not equal to `str2`
- D. `str1` and `str2` are identical objects

33. Which operator will return True when `num1 <= num2`?

- A. `num1` is greater than `num2`
 - B. `num1` is less than or equal to `num2`
 - C. `num1` is not equal to `num2`
 - D. `num1` equals `num2` only
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Assignment Operators

34. Which of the following is equivalent to `x += y`?

- A. `x = x - y`
- B. `x = x + y`
- C. `x = x * y`
- D. `x = y + y`

35. What will be the output of the following code?

```
a = 'India'
```

```
a *= 3
```

```
print(a)
```

- A. `'India3'`
- B. `'IndiaIndiaIndia'`
- C. `'India'`
- D. Error

36. Which assignment operator multiplies the current value of a variable by the right-hand operand?

- A. `*=`
- B. `/=`
- C. `%=`
- D. `+=`

Logical Operators

37. Which Python operator returns True if both operands are True?

- A. or
- B. and
- C. not
- D. ==

38. What is the output of the following code?

```
num1 = 10
```

```
num3 = 0
```

```
print(bool(num1 and num3))
```

- A. True
- B. False
- C. 0
- D. 1

39. Which logical operator reverses the logical state of its operand?

- A. and
- B. or
- C. not
- D. xor

Identity Operators

40. Which operator checks whether two variables point to the same memory location?

- A. ==
- B. is
- C. =
- D. !=

41. What does `var1 is not var2` evaluate to if `var1` and `var2` are pointing to the same object?

- A. True
- B. False
- C. Error
- D. None

Membership Operators

42. Which operator checks whether a value exists in a given sequence?

- A. is
- B. in
- C. not
- D. not in

43. What is the result of `10 not in [1,2,3,4]`?

- A. True
 - B. False
 - C. Error
 - D. 10
-

Operator Precedence

44. Which operator has the highest precedence in Python?

- A. +, -
- B. *, /, %, //
- C. **
- D. ==, !=, <, >

45. How will Python evaluate `20 + 30 * 40`?

- A. 2000
- B. 1220
- C. `50 * 40`
- D. 500

46. What will be the output of `20 - 30 + 40`?

- A. -10
- B. 30
- C. 20
- D. 0

47. Which of the following evaluates first in the expression `15.0 / 4 + (8 + 3.0)`?

- A. `15.0 / 4`
 - B. `8 + 3.0`
 - C. Addition of both results
 - D. Subtraction
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Statements and Input/Output

48. In Python, a statement is defined as

- A. A comment in the code
- B. A unit of code that the interpreter can execute
- C. A variable declaration
- D. An operator

49. Which function is used to take input from the user in Python?

- A. `print()`
- B. `input()`
- C. `scan()`
- D. `read()`

50. What is the datatype of the value returned by `input()` by default?

- A. int
- B. float

- C. str
- D. bool

51. Which function is used to display output to the user in Python?

- A. write()
- B. display()
- C. print()
- D. echo()

52. In the print() function, which parameter is used to change the separator between values?

- A. sep
- B. end
- C. space
- D. delim

53. What will be the output of the following code?

```
print("I'm", 16, "years old")
```

- A. I'm16years old
 - B. I'm 16 years old
 - C. "I'm 16 years old"
 - D. Error
-

54. What will be the output of the following code if the user enters 2?

```
num1 = input("Enter a number: ")  
  
num1 = num1 * 2  
  
print(num1)
```

- A. 4
- B. 22
- C. Error
- D. 2

55. Why does the program in Q54 produce 22 instead of 4?

- A. Because Python cannot multiply numbers
- B. Because input() returns a string by default
- C. Because the print statement is wrong
- D. Because Python does not allow arithmetic operations

56. How can we modify the program to display 4 when user enters 2?

- A. Convert input to integer using int()
- B. Convert input to string using str()
- C. Multiply input by 3
- D. Use print() twice

57. What is type conversion in Python?

- A. Changing the name of a variable
- B. Changing the value of a variable

- C. Changing the data type of a variable from one type to another
- D. Adding two variables

58. Which type of conversion happens when the programmer forces it?

- A. Implicit conversion
- B. Explicit conversion
- C. Logical conversion
- D. Automatic conversion

59. What will be the output of the following code?

```
x = 20.67
```

```
y = int(x)
```

```
print(y)
```

- A. 20.67
- B. 21
- C. 20
- D. 0

60. Which of the following functions converts an integer to a float?

- A. float(x)
- B. int(x)
- C. str(x)
- D. chr(x)

61. What will the following code produce?

```
priceIcecream = 25
```

```
priceBrownie = 45
```

```
totalPrice = priceIcecream + priceBrownie
```

```
print("The total is Rs." + totalPrice)
```

- A. The total is Rs.70
- B. Error
- C. The total is Rs.2545
- D. None

62. Which function is used to explicitly convert a string to an integer?

- A. str()
- B. int()
- C. float()
- D. ord()

63. What is implicit conversion (coercion) in Python?

- A. Conversion done automatically by Python without programmer instruction
- B. Conversion forced by the programmer
- C. Conversion using print() function
- D. Conversion using input() function

64. What will be the output of the following program?

```
num1 = 10  
  
num2 = 20.0  
  
sum1 = num1 + num2  
  
print(type(sum1))
```

- A. <class 'int'>
 - B. <class 'float'>
 - C. <class 'str'>
 - D. Error
-

Debugging in Python

65. What is debugging?

- A. Writing comments in a program
- B. Writing efficient code
- C. Identifying and removing errors from a program
- D. Running a program without print statements

66. Which type of error occurs when Python syntax rules are violated?

- A. Runtime error
- B. Syntax error
- C. Logical error
- D. Semantic error

67. Which type of error produces incorrect output but does not stop program execution?

- A. Syntax error
- B. Logical error
- C. Runtime error
- D. Type error

68. What will happen if you divide a number by zero in Python?

- A. Program will terminate with a runtime error
- B. Program will produce 0
- C. Program will produce infinity
- D. Program will ignore the statement

69. Which of the following is an example of a logical error?

- A. (10 + 12)
- B. (10 + 12/2) when calculating average of 10 and 12
- C. print("Hello"
- D. Dividing by zero

70. If a variable is used before assigning a value, what type of error occurs?

- A. Runtime error
- B. Logical error
- C. Syntax error
- D. No error

71. Which of the following is NOT a data type in Python?

- A. integer
- B. boolean
- C. decimal
- D. dictionary

72. What will be the type of variable returned by input() by default?

- A. int
- B. str
- C. float
- D. bool